



Cloud Storage for AI: Object, Block, and File Systems

A comprehensive guide for cloud engineers and AI practitioners building modern storage architectures

Why Storage Matters for AI

AI workloads demand **enormous datasets and model weights** that require thoughtful storage architecture. The storage type you select directly impacts **speed, scalability, and cost** - three critical factors that can make or break your AI infrastructure.

Cloud platforms offer multiple storage models, each with distinct characteristics. Engineering teams must carefully balance performance requirements against durability needs and budget constraints. As models grow larger and datasets expand, your storage decisions become increasingly consequential.



Three Main Storage Types

Object Storage

Highly scalable storage for unstructured data

- Infinite capacity potential
- Cost-effective for large datasets
- API-driven access patterns

Block Storage

Fast disk volumes for compute instances

- Low-latency performance
- Instance-attached resources
- Similar to physical SSDs/HDDs

File Storage

Shared access via filesystem protocols

- Multi-server mounting capability
- Familiar directory structure
- Team collaboration-friendly

These storage types are often combined in hybrid architectures to address [different AI workload needs](#) throughout the machine learning lifecycle.

Object Storage – Overview

Object storage treats data as discrete **objects** consisting of:

- The data itself (binary blob)
- Metadata (properties and attributes)
- Unique identifier (global namespace)

Popular implementations include [AWS S3](#), [Google Cloud Storage](#), and [Azure Blob Storage](#). These services offer virtually unlimited scalability at relatively low cost with exceptional durability guarantees (often 11 9's).

Unlike traditional file systems, object storage is accessed via REST APIs rather than filesystem paths, making it ideal for **datasets, logs, and model checkpoints**.



Object Storage – AI Use Cases

Training Dataset Storage

Massive collections of images, text, video, and other unstructured data used for model training. Object storage's unlimited capacity makes it perfect for growing datasets.

Model Weight Preservation

Save trained model weights, experiment artifacts, and checkpoints durably. Essential for reproducing results and deploying models to production environments.

Data Lake Implementation

Centralized repository allowing data scientists to store, discover, and preprocess structured and unstructured data at scale using tools like Spark, Ray, or custom ETL pipelines.

Backup & Archival

Long-term, cost-effective storage for AI artifacts with infrequent access patterns. Many object stores offer archival tiers with significantly reduced pricing.

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graph LR; A[Object Storage] --> B[Preprocessing]; B --> C[Training Cluster];
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Object Storage

Preprocessing

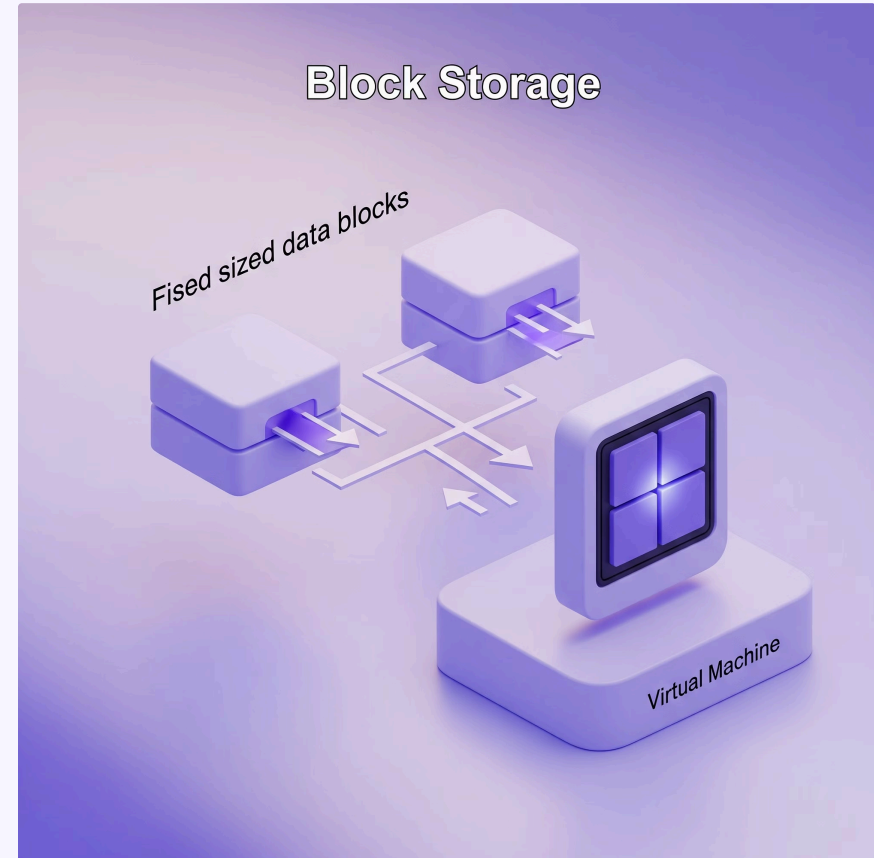
Training Cluster

Block Storage – Overview

Block storage functions as a **virtual hard drive** attached to your compute instances, providing:

- **Low latency access** - microsecond response times
- **High throughput** - hundreds of MB/s to GB/s
- **Instance binding** - typically attached to specific VMs
- **Familiar interface** - appears as a local disk volume

Examples include [AWS Elastic Block Store \(EBS\)](#), [Google Persistent Disk](#), and [Azure Managed Disks](#). These services offer various performance tiers based on underlying technology (SSD vs HDD) and specialized optimizations.



Block Storage – AI Use Cases



Dataset Preprocessing

High-performance I/O for data transformation pipelines, feature engineering, and ETL processes that require rapid access to intermediate results.



Training Jobs

ML workloads requiring fast, local storage for mini-batches, especially when processing data that doesn't fit in RAM or when using frameworks with specific I/O patterns.



Feature Stores

Hosting databases and caching layers for ML feature stores, where low latency is critical for both training and inference applications.

Block storage excels when your AI workflow needs [high-speed, local disk access](#) with performance characteristics similar to direct-attached storage. It's particularly valuable for compute-intensive tasks with high I/O requirements that don't need multi-server access patterns.

File Storage – Overview

File storage provides [shared access via standard filesystem protocols](#) (NFS, SMB) with these key characteristics:

- Simultaneously mountable across multiple compute instances
- Hierarchical directory structure familiar to all developers
- Granular permissions and access controls
- Traditional POSIX-compliant interface

Major cloud offerings include [Amazon EFS](#), [Azure Files](#), and [Google Cloud Filestore](#). These managed services eliminate the complexity of setting up and maintaining traditional network-attached storage while preserving its familiar interface.



File Storage – AI Use Cases

1 Team Collaboration

Enables multiple ML engineers to access the same datasets, code, and models simultaneously. Particularly valuable for research teams iterating on shared experiments.

2 Multi-Server Model Serving

Hosts pre-trained models across multiple inference servers from a single, consistent location. Simplifies deployment and ensures all servers access identical model artifacts.

3 Interactive Development

Powers collaborative environments like JupyterHub or MLflow where multiple users need concurrent access to shared resources, notebooks, and experiment tracking.

File storage shines in scenarios requiring [shared access patterns across compute resources](#). When multiple nodes need to read and write to the same filesystem with consistent semantics, file storage provides the necessary abstraction while maintaining acceptable performance.

Comparing Storage Types for AI Workloads

Attribute	Object Storage	Block Storage	File Storage
Scalability	Virtually unlimited	Limited by volume size	High but with limits
Performance	Moderate throughput, high latency	High throughput, low latency	Variable, network-dependent
Cost	Very low (¢ per GB)	Moderate (10¢-50¢ per GB)	Higher (20¢-\$1+ per GB)
Multi-instance	Via API only	Single instance only*	Multiple instances
Access Pattern	API/HTTP requests	Direct disk I/O	Filesystem operations
Ideal for	Datasets, archives, checkpoints	High-speed training, preprocessing	Shared workspaces, collaboration

** Some cloud providers offer multi-attach block storage with limitations*

Sophisticated AI infrastructure typically combines all three storage types strategically, leveraging their complementary strengths throughout the ML lifecycle.

Key Takeaways

Your storage architecture fundamentally shapes the efficiency and capabilities of your AI pipelines:

1 Storage is a Strategic Decision

Storage choices impact performance, cost, and flexibility across your entire ML workflow. Design deliberately.

2 Layer Appropriately

Use **object storage** for datasets and archives, **block storage** for high I/O training, and **file storage** for collaboration.

3 Optimize for Workload

Match storage characteristics to specific AI tasks. Different phases of the ML lifecycle have different storage requirements.



Effective AI infrastructure engineers mix storage types strategically to balance performance, cost, and accessibility needs.